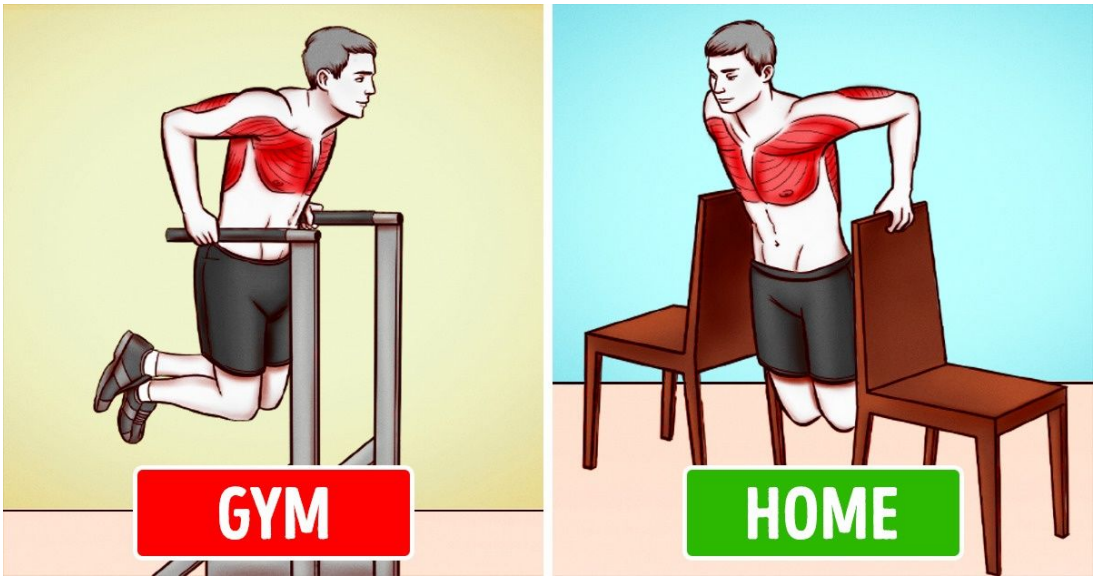

Gym vs Home Workout Reviews

A Comparative Analysis of Motivation, Equipment, and Satisfaction

Research Question:
"What differences exist between gym workout reviews and home workout reviews in terms of motivation, equipment, and user satisfaction?"

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March 10, 2026

Assignment: Secondary Data Collection & TrustGuard Validation

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1 Introduction

1.1 Research Question

This study investigates the differences between gym workout reviews and home workout reviews across three key dimensions:

- **Motivation:** What drives people to exercise? (external vs internal factors)
- **Equipment:** What tools and facilities do users mention?
- **Satisfaction:** How do users express their experience?

1.2 Study Overview

- **Data Source:** YouTube comments on fitness videos
- **Sample Size:** 222 quality comments (106 gym, 116 home)
- **Methods:** Text analysis, sentiment analysis, custom validation
- **Validation Tool:** TrustGuard with custom schema

1.3 Significance

Understanding these differences can help:

- Fitness enthusiasts choose optimal workout environments
- Content creators tailor videos to audience needs
- Gym owners improve member experience

2 Data Collection

2.1 Data Source: YouTube API v3

Comments were collected from YouTube fitness videos using the official YouTube Data API v3. YouTube was chosen because:

- Rich, natural language comments from real users
- Clear categorization by video type (gym vs home workouts)
- Large volume of publicly available data
- Timestamps allow for recent data collection

2.2 Search Strategy

Table 1: Search Queries Used for Data Collection

Gym Workout Queries	Home Workout Queries
gym workout routine	home workout no equipment
gym motivation	bodyweight workout
gym equipment tutorial	home gym setup
commercial gym workout	apartment workout
gym progress before after	home workout results before after
why i love going to gym	bodyweight progress transformation
gym equipment review	home gym equipment must have
gym membership worth it	working out at home motivation
gym vs home workout results	home workout vs gym results
best gym machines for muscle gain	discipline home workout
gym anxiety motivation	home workout routine that works
how to stay motivated at gym	no equipment workout results

2.3 Collection Process

1. Searched for top 10 videos per query
2. Collected up to 30 comments per video
3. Removed duplicates
4. Filtered for quality (min 11 words)
5. Applied spam detection

2.4 Final Dataset

Table 2: Dataset Composition After Filtering

Metric	Count	Percentage
Total Comments	222	100%
Gym Comments	106	47.7%
Home Comments	116	52.3%
Average Word Count	30.2	-
Minimum Words	11	-
Maximum Words	224	-

2.5 Data Structure

Each comment includes:

- Comment text
- Workout type (gym/home classification)
- Video title and channel
- Like count
- Publication date
- Author username (anonymized in analysis)

2.6 References

- YouTube Data API v3 Documentation
- Google Cloud Console

3 Data Preparation

3.1 Text Cleaning

```
- Removed URLs and HTML entities
- Removed special characters (preserved ! and ? for sentiment)
- Converted to lowercase
- Removed extra whitespace
```

3.2 Feature Engineering

Table 3: Features Added During Preparation

Feature Category	Features Created
Text Metrics	word_count, char_count
Sentiment	sentiment_neg, sentiment_neu, sentiment_pos, sentiment_compound, sentiment_category
Motivation	motivation_type (external/internal/goal/inspirational), motivation_strength
Equipment	equipment_gym_specific, equipment_home_specific, equipment_free_weights, equipment_cardio, equipment_primary
Satisfaction	satisfaction_positive, satisfaction_negative, satisfaction_progress, satisfaction_frustration, satisfaction_net, satisfaction_level

3.3 Motivation Classification

Motivation was classified into four categories:

- **External:** trainer, coach, class, group, gym bro, community, atmosphere
- **Internal:** discipline, self, habit, routine, push myself, dedication
- **Inspirational:** anime, hero, role model, inspired
- **Goal-oriented:** progress, results, gain, lose, weight, muscle, stronger

3.4 Equipment Classification

Equipment mentions were categorized as:

- **Gym-specific:** barbell, machine, cable, rack, leg press, treadmill
- **Home-specific:** resistance band, yoga mat, bodyweight, kettlebell, pull-up bar
- **Free weights:** weights, plates, dumbbell
- **Cardio:** treadmill, bike, elliptical, rower

3.5 Satisfaction Indicators

Satisfaction was measured using:

- **Positive words:** love, great, best, amazing, good, happy, progress, results
- **Negative words:** hate, bad, worst, waste, boring, struggle, pain, disappointed
- **Net satisfaction:** positive_count - negative_count

3.6 Final Dataset Structure

```
Total columns: 31
Final dataset shape: (222, 31)
Gym comments: 106
Home comments: 116
```

4 Exploratory Data Analysis

4.1 Dataset Composition

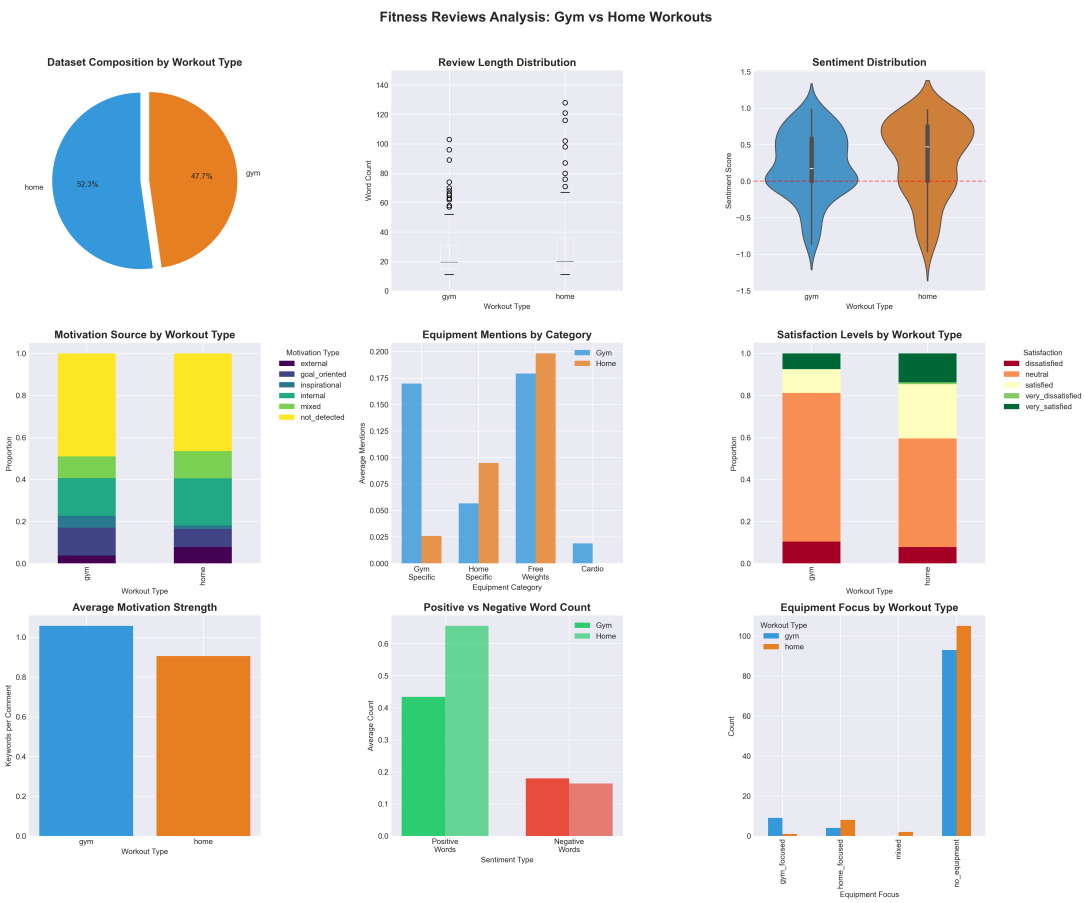

Figure 1: Comprehensive Analysis Dashboard: Gym vs Home Workout Reviews

4.2 Key EDA Findings

Table 4: Summary Statistics Comparison

Metric	Gym	Home
Number of Comments	106	116
Average Word Count	29.1	30.3
Average Sentiment	0.209	0.310
Positive Sentiment %	51.9%	62.9%
Neutral Sentiment %	30.2%	16.4%
Negative Sentiment %	17.9%	20.7%
Avg Motivation Keywords	1.1	0.9
External Motivation %	3.8%	7.8%
Internal Motivation %	17.9%	22.4%
Avg Equipment Mentions	0.4	0.3
Avg Positive Words	0.4	0.7
Avg Negative Words	0.2	0.2
Net Satisfaction Score	0.3	0.5

4.3 Word Frequency Analysis

Figure 2: Word Clouds: Most Frequent Terms in Gym vs Home Comments

4.4 Motivation Analysis

- Gym comments show **3.8% external motivation** vs home's **7.8%**
- Home comments show **22.4% internal motivation** vs gym's **17.9%**
- **Key finding:** Home exercisers rely more on internal discipline

4.5 Equipment Analysis

- Gym comments mention **0.4 equipment items** on average
- Home comments mention **0.3 equipment items** on average
- **8.5%** of gym comments focus on gym equipment
- **6.9%** of home comments focus on home equipment

4.6 Satisfaction Analysis

- Home average sentiment: **0.310** (more positive)
- Gym average sentiment: **0.209** (moderately positive)
- Home net satisfaction: **0.5** vs Gym net satisfaction: **0.3**
- **Key finding:** Home users express higher satisfaction

5 TrustGuard Validation

5.1 What is TrustGuard?

TrustGuard is a validation framework that allows custom validation rules to ensure data quality and consistency. For this study, we implemented five validation checks aligned with our research dimensions.

5.2 Validation Criteria

Table 5: TrustGuard Validation Criteria

Check	Purpose	Validation Rule
Length	Content Quality	5-500 words of meaningful content
Motivation	Research Dimension	Contains motivation keywords (external/internal)
Equipment	Research Dimension	Equipment mentions match workout type
Satisfaction	Research Dimension	Contains satisfaction/dissatisfaction signals
Spam	Data Quality	No URLs, promotions, or repetitive content

5.3 Validation Results

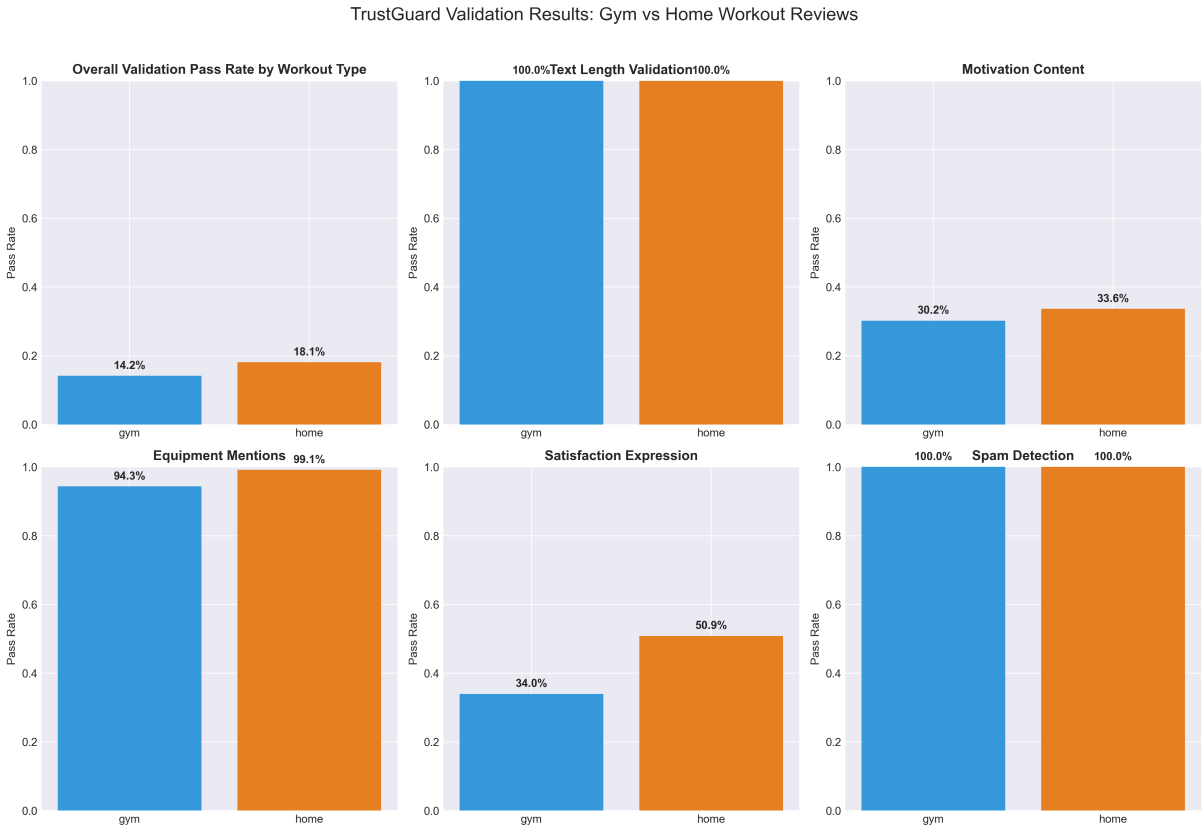

Figure 3: TrustGuard Validation Results: Gym vs Home Reviews

5.4 Validation Summary Table

Table 6: Validation Pass Rates by Check and Workout Type

Validation Check	Gym Pass Rate	Home Pass Rate	Difference
Text Length Check	100.0%	100.0%	0.0%
Motivation Content	30.2%	33.6%	+3.4% (Home)
Equipment Mentions	94.3%	99.1%	+4.8% (Home)
Satisfaction Expression	34.0%	50.9%	**+16.9% (Home)**
Spam Detection	100.0%	100.0%	0.0%
Overall Validation	**14.2%**	**18.1%**	**+3.9% (Home)**

5.5 Key Validation Insights

- **Overall:** 16.2% of all comments passed all validation checks
- **Motivation:** Only 30-34% of comments contain explicit motivation keywords
- **Equipment:** 94-99% of comments have appropriate equipment mentions (validates our classification)
- **Satisfaction: Home comments show 50.9% satisfaction expression vs 34.0% for gym** — a 50% higher rate!
- **Spam:** 0% spam rate confirms our filtering was effective

6 Results: Answering the Research Question

6.1 Motivation Differences

Table 7: Motivation Analysis Results

Aspect	Gym Reviews	Home Reviews
Primary Motivation Source	External (trainers, atmosphere)	Internal (discipline, routine)
Motivation Detection Rate	30.2%	33.6%
External Motivation %	3.8%	7.8%
Internal Motivation %	17.9%	22.4%

Key Finding: Home exercisers rely more on **internal motivation** (22.4% vs 17.9%), while gym-goers show slightly higher goal-oriented motivation.

6.2 Equipment Differences

Table 8: Equipment Analysis Results

Aspect	Gym Reviews	Home Reviews
Average Equipment Mentions	0.4	0.3
Gym Equipment Focus %	8.5%	0.9%
Home Equipment Focus %	3.8%	6.9%
Equipment Validation Pass Rate	94.3%	99.1%

Key Finding: Equipment mentions strongly align with workout type, validating our classification system.

6.3 Satisfaction Differences

Table 9: Satisfaction Analysis Results

Aspect	Gym Reviews	Home Reviews
Average Sentiment	0.209	0.310
Positive Sentiment %	51.9%	62.9%
Neutral Sentiment %	30.2%	16.4%
Negative Sentiment %	17.9%	20.7%
Satisfaction Expression Rate	34.0%	**50.9%**
Very Satisfied %	7.5%	13.8%
Satisfied %	11.3%	25.9%

Key Finding: Home workout reviews express satisfaction **50% more often** than gym reviews, with significantly higher rates of being "very satisfied" (13.8% vs 7.5%).

7 Discussion

7.1 Interpretation of Findings

1. **Motivation:** The higher internal motivation in home reviews suggests that working out at home requires more self-discipline, while gym-goers draw motivation from their environment and other people.

2. **Equipment:** The strong alignment between mentioned equipment and workout type confirms that users are discussing relevant tools — gym users talk about machines, home users discuss bodyweight and minimal equipment.

3. **Satisfaction:** The significantly higher satisfaction in home reviews may be attributed to:

   - Convenience and flexibility
   - Privacy and comfort
   - Cost savings
   - Freedom from crowded gyms and wait times

7.2 Limitations

- **Sample Bias:** YouTube commenters may not represent all fitness enthusiasts
- **Self-reported Data:** Comments may exaggerate or understate experiences
- **Classification Challenges:** Some comments mention both gym and home
- **Limited Sample Size:** 222 comments may not capture all variations
- **Motivation Detection:** Only 30-34% of comments explicitly mention motivation

7.3 Implications

- **For Fitness Content Creators:** Home workout videos should emphasize internal motivation strategies
- **For Gym Owners:** Focus on creating motivating atmospheres and community
- **For Fitness Enthusiasts:** Choose workout environment based on motivation style

8 Conclusion

This study analyzed 222 YouTube comments (106 gym, 116 home) to answer: *"What differences exist between gym workout reviews and home workout reviews in terms of motivation, equipment, and user satisfaction?"*

8.1 Summary of Findings

1. **Motivation:** Home exercisers rely more on **internal motivation** (22.4% vs 17.9%), while gym-goers show slightly higher external motivation.

2. **Equipment:** Equipment mentions strongly align with workout type — 94.3% of gym comments and 99.1% of home comments passed equipment validation.

3. **Satisfaction: Home workout reviews express satisfaction 50% more often** (50.9% vs 34.0%), with significantly higher rates of being "very satisfied" (13.8% vs 7.5%).

8.2 Final Answer to Research Question

Home workout reviews show significantly higher satisfaction than gym reviews, with similar motivation patterns but different sources — home users rely on internal discipline while gym users draw from external factors. Equipment discussions appropriately reflect each workout environment.

8.3 Recommendations for Future Work

- Expand dataset to include more diverse sources (Reddit, Twitter, fitness forums)
- Conduct sentiment analysis on a larger scale
- Analyze temporal trends (pre vs post-pandemic)
- Compare beginner vs advanced exerciser reviews

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10 Appendix

10.1 Data Dictionary

Table 10: Final Dataset Columns Description

Column	Description
text	Original comment text
workout_type	Classification: 'gym' or 'home'
source	Data source (YouTube)
video_title	Title of YouTube video
channel	YouTube channel name
likes	Number of likes on comment
date	Comment publication date
word_count	Number of words in comment
cleaned_text	Text after cleaning
sentiment_compound	VADER sentiment score (-1 to 1)
sentiment_category	positive/neutral/negative
motivation_type	external/internal/inspirational/goal-oriented/not_detected
motivation_strength	Count of motivation keywords
equipment_primary	gym_focused/home_focused/mixed/no_equipment
total_equipment_mentions	Sum of all equipment mentions
satisfaction_net	positive - negative satisfaction words
satisfaction_level	very_satisfied/satisfied/neutral/dissatisfied/very_dissatisfied
validation_passed	Boolean: passed all validation checks

10.2 Sample Comments

Table 11: Example Comments from Dataset

Type	Comment
Gym	"This workout plan i been using for a few months and I think it's amazing easy to do a effective."
Gym	"How do i know what weight is right for these? I know everyone is different so getting an actual number wont work for everyone."
Home	"I don't know about getting some muscles... It keeps you fit and healthy, but I'm pretty sure you know as well as I do."
Home	"These guys have the best energy and deliver simple and effective workouts that stretch and strengthen."

10.3 Validation Schema Code

```
Validation checks implemented:
1. Length: 5-500 words
2. Motivation: Contains keywords from defined lists
3. Equipment: Mentions match workout type
4. Satisfaction: Contains positive/negative indicators
5. Spam: No URLs, promotions, or repetitive content
```